

DOMAIN KNOWLEDGE MAP

Working in Energy Products & Tariffs

1KOMMA5°

Everything you need to know to be genuinely competent — and therefore confident — in a European dynamic-tariff product team.
English content with short Turkish explanations.

Prepared for: **Ali Mert Özdemir**

Role: Internship Product Management — Energy Products & Tariffs (m/f/d)

Locations: Berlin / Hamburg / München / Remote · Duration: min. 6 months, ideally full-time

Interview: 45-minute video call (1st round)

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0. Read this first — what the job actually is

The job title says "Product Management". The job content is **energy-market product ownership inside a regulated retail business**. This is not a data-engineering role, and it is not a strategy role. It sits in the middle of four functions and translates between them:

Function	What they give you	What they need from you
Regulation / Legal	New laws, BNetzA rulings, deadlines	Translation into concrete product and system decisions
Trading / Energy Ops	Spot prices, hedging, forecast, imbalance cost	Tariff structures they can actually source and hedge
Operations / Billing	Market communication, switching, invoicing	Process fixes that reduce failures and manual work
Commercial / Growth	Conversion, churn, competitor pricing	Products that sell and still carry margin

TR: Bu rol veri mühendisliği değil. İşin özü: enerji piyasası kurallarını ve fiyat mekaniğini anlayıp, bunları satılabilir ve faturalanabilir bir tarif ürününe çevirmek. Dört ekibin (regülasyon, trading, operasyon/ faturalama, ticari) ortak dilini konuşmak zorundasın.

The five things the job description literally asks for

1. **Own well-scoped initiatives** — analysis → recommendation → implementation, with accountability for the outcome.
2. **Work on dynamic electricity tariff products** across core markets — improvements, feature rollouts, customer experience.
3. **Support regulatory product launches** — the JD names **Direktvermarktung** explicitly.
4. **Analyse and improve Marktkommunikation processes** — "make our energy retail engine faster and more reliable".
5. **Monitor KPIs and present** to Product Managers and stakeholders.

Sections 2.7 (Direktvermarktung) and 2.10 (Marktkommunikation) of this document are therefore the two highest-leverage topics in it. Most candidates for an internship cannot talk about Marktkommunikation at all. If you can, you are immediately in a different category.

TR: İlanda ismi geçen iki teknik konu **Direktvermarktung** ve **Marktkommunikation**. Bu dokümandaki 2.7 ve 2.10 bölümleri bu yüzden en kritik. Staj adaylarının neredeyse hiçbiri Marktkommunikation'ı anlatamaz — sen anlatabilirsən doğrudan ayırırsın.

The uncomfortable part, stated up front. Two real risks, neither of them technical:

- **German.** The profile says "excellent written and spoken English *and German*". Marktkommunikation, EnWG, EEG, BNetzA rulings and DSO correspondence are all in German. If your German is not conversational, expect this question and answer it honestly with a plan (current level, what you are doing about it, and the fact that you can read energy-regulatory German — which is true if you work through Section 7 of this document).
- **Work permit.** The invitation asks for your valid work permit including **Zusatzblatt**, plus residence permit. Have scans ready *before* the call. Know your own permitted working hours as a student. Do not improvise this answer.

TR: İki gerek risk teknik deęil: (1) **Almanca** — ilanda "excellent" yazıyor, MaKo ve mevzuat Almanca. Dürüst seviye + somut plan söyle. (2) **alıřma izni** — mailde Zusatzblatt dahil belge istiyorlar; görüřmeden önce taratıp hazır tut ve haftalık izinli alıřma saatini net bil.

1. The company — understand the business model, not the marketing

1.1 Three revenue engines

1KOMMA5° is not a solar installer and not an energy supplier. It is deliberately all three layers of the home-energy stack at once, because each layer monetises the previous one:

Layer	What it is	Economics
1. Hardware + installation	One-stop-shop: PV, battery, heat pump, wallbox, installed by acquired regional craft businesses ("Meisterbetriebe")	High revenue, low margin, cyclical, capital-intensive. This is the customer-acquisition engine.
2. Heartbeat AI (software)	Home energy management + control layer; a flat software fee; now also opened to <i>existing third-party</i> systems	Recurring, high-margin, asset-light. This is the equity story.
3. Energy retail + flexibility	Dynamic tariff supply, aggregation of the installed base into a virtual power plant (VPP) trading on wholesale and flexibility markets	Recurring margin per customer per year, scales with the installed base. This is your team.

The strategic logic: layer 1 buys the customer, layer 2 gives control of the assets, layer 3 monetises that control every day for 20 years. Layer 3 is what turns a hardware company into a software-and-energy company — which is exactly what matters ahead of an IPO.

TR: Üç katman: (1) donanım+kurulum = müşteri kazanma motoru, düşük marj; (2) Heartbeat yazılımı = tekrarlayan yüksek marj; (3) enerji tedariği + esneklik ticareti = her gün gelir üreten katman ve senin ekibin. Şirketin değerlemesi 1. katmandan çok 2. ve 3. katmana bağlı — bu yüzden tarif ekibi stratejik merkezde.

1.2 Numbers worth holding in your head

Metric	Value	Why it matters
Revenue 2024	≈ €520M (+36% organic)	Grew while the German solar market shrank
Revenue outlook 2026	≈ €660M, EBITDA-positive	Shift from growth-at-any-cost to profitable scale
Latest raise	€150M pre-IPO round	IPO is the near-term corporate goal
Heartbeat AI investment	> €100M planned 2025-2027	Software/energy is where the money is going
VPP flexibility	500 MW (2025) → 1 GW shiftable load (2026)	Claimed largest residential VPP in Europe
Connected systems	> 100,000 systems; > 120,000 customers	The base your tariff products are sold into
Scale	3,000+ people, 80+ locations, Finland → Australia	"Core markets" in the JD = multi-country, not just DE
2030 target	1.5 million households on renewables	The number the whole company is steered by

Leadership	Philipp Schröder (CEO, ex-Tesla DE, ex-sonnen), Micha Grüber	Founder is an energy-market operator, not a generalist
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TR: Bu tablodaki 3-4 sayıyı ezberle (520M ciro, 1 GW esnek kapasite, 100k+ bağlı sistem, 2030'da 1,5M hane). Görüşmede "şirketi anlamış" izlenimi bu sayılarla verilir, genel övgüyle değil.

1.3 The product that explains the whole strategy: "free electricity"

In Sweden, 1KOMMA5° launched **Dynamic Pulse**: customers whose PV *and* battery are connected to Heartbeat get electricity at **0 cent/kWh for 15,000 kWh per year**. You should be able to explain why this is not a giveaway:

- The customer surrenders **control** of the battery (and increasingly the heat pump / EV) to the aggregator.
- The aggregator monetises that control: spot arbitrage (charge cheap / discharge expensive), intraday trading, balancing-market capacity, avoided grid fees, and imbalance-portfolio smoothing.
- If the flexibility value captured per customer per year exceeds the wholesale cost of that customer's energy, giving the energy away is rational — and it is an extremely strong acquisition and retention weapon.
- It also locks the customer to the platform: leaving means losing free electricity. Churn drops.

The follow-up question you should be ready for: "*would this work in Germany?*" A defensible answer names the blockers rather than guessing: German retail price is ~two-thirds non-commodity (grid fees, levies, taxes, VAT) which you cannot zero out; battery-only flexibility earns less where spot spreads are narrower; and §14a curtailment plus metering coverage limit how much flexibility is actually addressable. So the German version is more likely to be a *deep discount on the commodity component* than a headline zero.

TR: İsveç'teki "bedava elektrik" ürünü hediye değil: müşteri bataryanın kontrolünü veriyor, şirket bu esnekliği piyasada paraya çeviriyor. Almanya'da aynısı zor — çünkü faturanın ~%60'ı şebeke bedeli, vergi ve harçlar; onları sıfırlayamazsın. Bu cevabı verebilmek, ürün mantığını anladığını gösterir.

1.4 Competitive map — know who you are compared against

Player	Model	Where 1KOMMA5° differs
Tibber	Dynamic tariff + app + hardware ecosystem, monthly fee, "Pulse" optical meter reader	Tibber is software-first without owning installation; 1K5 owns the installer network
Octopus Energy	Large-scale retail + Kraken platform licensing; day-ahead prices published a day ahead (Agile)	Octopus is a platform/retail giant; 1K5 is asset-and-flex centric
aWATTar	Lowest base fee, spot passthrough, technical/enthusiast audience	Price-led niche, no hardware
Ostrom	Pure passthrough: no markup on the spot price, higher monthly fee	Radical transparency positioning
Rabot Charge	Dynamic tariff optimised around EV charging	Single-device focus
Enpal	Closest structural rival in Germany: vertically integrated PV + battery + heat pump + tariff + VPP	Head-on competitor on the same bundle logic
sonnen	Battery manufacturer with sonnenCommunity VPP	Hardware-led VPP, narrower stack

E.ON / EnBW / Vattenfall / Stadtwerke	Incumbents, now legally obliged to offer dynamic tariffs	Huge base, slow product cycles, weak device control
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Honest note on positioning: independent German comparisons generally conclude that 1KOMMA5° is *not* the cheapest standalone dynamic tariff and only makes economic sense as part of the PV + storage + wallbox bundle. Knowing this — and framing it as the central product tension your team has to solve — is far stronger than pretending it is the cheapest.

TR: Bağımsız karşılaştırmalar 1KOMMA5°'yi "tek başına tarif olarak pahalı, ama paketle birlikte mantıklı" diye konumlandırıyor. Bunu bilip "işte ekibin çözmesi gereken asıl gerilim bu" diye çerçevelemek, şirketi savunmaya çalışmaktan çok daha güçlü durur.

2. Energy domain — the core 60%

2.1 Market roles: who is allowed to do what

Almost every product question in European energy retail eventually becomes "which role do we hold, and what does that role permit?". Learn these:

Role (DE)	EN	Function
Erzeuger	Generator	Produces electricity (utility-scale or rooftop)
Übertragungsnetzbetreiber (ÜNB)	TSO	High-voltage transmission, system balance. Germany has four: 50Hertz, Amprion, TenneT DE, TransnetBW
Verteilnetzbetreiber (VNB)	DSO	Distribution grid, connection, grid fees, §14a control. ~850 of them in Germany — a major source of process friction
Lieferant	Supplier / retailer	Sells energy to the end customer, holds the supply contract, invoices
Bilanzkreisverantwortlicher (BKV)	Balance responsible party	Commits to a schedule and pays for deviations. The financial heart of retail
Messstellenbetreiber (MSB)	Metering point operator	Installs and operates the meter. gMSB = default (usually the DSO), wMSB = competitive third party
Direktvermarkter	Direct marketer	Sells a generator's output on the market, carries the forecast risk
Aggregator	Aggregator / VPP operator	Pools small assets into a tradable, controllable unit
Grundversorger	Default supplier	Legally obliged local supplier of last resort; the expensive fallback tariff
Bundesnetzagentur (BNetzA)	Federal regulator	Writes the rulebooks (GPKE, MaBiS, §14a determinations); its "Beschlüsse" create product deadlines

1KOMMA5° holds several of these simultaneously: installer, HEMS provider, **Lieferant**, **BKV**, **Aggregator**, and increasingly **Direktvermarkter**. That combination is the actual moat.

TR: Enerji piyasasında her şey "rol"e bağlı. Lieferant (tedarikçi), BKV (denge grubu sorumlusu), MSB (sayaç işletmecisi), Direktvermarkter (direkt satış), Aggregator (havuzlayıcı). 1KOMMA5° bunların çoğunu aynı anda taşıyor — asıl rekabet avantajı bu. BNetzA = düzenleyici kurum, kural kitaplarını o yazıyor.

2.2 Wholesale markets — where the price in your tariff comes from

A dynamic tariff is a retail wrapper around a wholesale price. You must know the sequence:

1. **Forward / futures market (EEX)** — years, quarters, months ahead. Used for hedging fixed-price tariffs.
2. **Day-ahead auction (EPEX SPOT)** — the reference price for almost every dynamic tariff. Bids close at **12:00 CET** for the following day; results are published early afternoon. Since **1 October 2025** the German day-ahead auction clears in **15-minute market time units (MTU)** — **96 prices per day** instead of 24. It is a *uniform-price* (single clearing price) auction, coupled across Europe via SDAC.
3. **Intraday** — continuous trading up to shortly before delivery, plus intraday auctions (IDA). This is where forecast errors get corrected. Coupled via SIDC.

4. **Balancing / Regelleistung** — TSO-procured reserves: **FCR** (Primärregelleistung), **aFRR** (Sekundärregelleistung), **mFRR** (Minutenreserve). Aggregated batteries can prequalify and earn capacity + energy payments here.
5. **Redispatch** — TSO-instructed generation shifts to relieve grid congestion. Redispatch 2.0 obligations apply from 100 kW upward, so single households are out of scope, but pooled assets are strategically relevant.

The 15-minute switch is a product event, not a technicality. 96 MTUs/day means: sharper price peaks and troughs, more arbitrage value for a battery, better matching of PV shape, more data volume (35,040 intervals/year per meter), and billing logic that must be quarter-hour native. Any tariff product built on hourly assumptions is now structurally mispriced. Expect this to come up.

TR: Sıra şu: vadeli piyasa (EEX) → gün-öncesi ihale (EPEX, saat 12:00 kapanış) → gün-içi (intraday) → dengeleme (FCR/aFRR/mFRR) → redispatch. **1 Ekim 2025'ten beri Alman gün-öncesi piyasası 15 dakikalık dilimlerle işliyor: günde 96 fiyat.** Bu, batarya için daha çok arbitraj fırsatı ve faturalama mantığında çeyrek-saat zorunluluğu demek. Ürün açısından kritik bir değişiklik.

Price formation you should be able to explain in 30 seconds

Merit order: plants are dispatched cheapest-marginal-cost first (renewables ≈ 0 → nuclear/lignite → coal → gas → oil). The *last* plant needed sets the price for everyone. So gas often sets the price even when renewables dominate volume. When renewable supply exceeds inflexible demand, prices go **negative** — producers pay to feed in. Negative prices are the single most important phenomenon for your product: they are simultaneously the reason dynamic tariffs are attractive, the reason feed-in support is being withdrawn, and the reason flexibility has value.

TR: Merit order = en ucuz santral önce devreye girer, fiyatı *son devreye giren* santral belirler. Bu yüzden yenilenebilir çok olsa bile fiyatı sık sık gaz belirler. Arz talebi aştığında fiyat **negatif** düşer. Negatif fiyatlar bu için merkezinde: dinamik tarifi cazibesi, EEG desteğinin geri çekilmesi ve esnekliğin değeri hep buradan geliyor.

2.3 Anatomy of a German retail price (2026)

This is the single most useful table in the document, because it tells you how much of the bill a dynamic tariff can actually influence.

Component	Approx. 2026	Dynamic?
Household average total price	≈ 37.2 ct/kWh (2025: 39.3)	—
Procurement + sales + margin	$\approx 41\%$ (≈ 15.4 ct)	Yes — this is the part that follows the spot price
Grid fees (Netzentgelte)	$\approx 24.8\%$ (≈ 9.26 ct)	Only via §14a Module 3 (time-variable)
Taxes, levies, surcharges	$\approx 33.9\%$ (≈ 12.61 ct)	No
· Electricity tax (Stromsteuer)	2.05 ct	No
· §19 StromNEV levy	1.56 ct	No
· Offshore grid levy	0.94 ct	No
· KWKG levy (cogeneration)	0.45 ct	No
· Concession fee + 19% VAT	remainder	No (VAT scales with the total)

The product conclusion: only roughly 40% of the bill is exposed to the wholesale price. A customer who shifts 30% of consumption into the cheapest quarter-hours does not cut their bill by 30% — they cut it by 30% of $\sim 40\%$, minus fees. Anyone in a tariff team who cannot do this arithmetic will over-promise savings, and over-promised savings are the #1 driver of churn and regulatory complaints in this segment.

Which is exactly why **§14a Module 3 (time-variable grid fees)** matters so much: it puts a second $\sim 25\%$ of the bill in play. Spot + time-variable grid fee = a real two-signal optimisation. This is where dynamic tariff products are heading, and it is a great thing to bring up unprompted.

TR: Faturanın sadece $\sim 40\%$ 'ı borsa fiyatına bağlı. 25% şebeke bedeli, 34% vergi/harç. Yani müşteri tüketiminin 30% 'unu ucuz saatlere kaydırırsa faturası 30% değil, kabaca $30\% \times 40\%$ kadar düşer. Bu aritmetiği bilmeyen biri tasarruf vaadini şişirir — ve şişirilmiş vaat bu sektörde 1 numaralı müşteri kaybı sebebi. §14a Modül 3 (zaman değişkenli şebeke bedeli) bu yüzden çok önemli: faturanın ikinci 25% 'ini de optimize edilebilir hale getiriyor.

2.4 Dynamic tariffs as a product

Legal base: since **1 January 2025**, §41a EnWG obliges every German electricity supplier to offer a dynamic tariff. This turned a differentiator into a commodity feature — so competition moved to *optimisation quality, hardware integration and transparency*, not to the existence of the product.

Price build-up of a dynamic tariff

Customer price per 15 min = Day-ahead price (MTU) + supplier markup (ct/kWh) + grid fee + levies + taxes + VAT, plus a **monthly base fee** (Grundgebühr).

Business model	Mechanics	Trade-off
Pure passthrough	0 markup on spot; revenue purely from the base fee (Ostrom-style)	Maximum trust, zero margin flexibility, no buffer for imbalance cost
Markup + low base fee	Small ct/kWh markup, minimal monthly fee (aWATTar-style)	Margin scales with volume; attractive to low-consumption customers
Ecosystem-bundled	Tariff attached to hardware + HEMS; margin comes from flexibility and the bundle, not the ct/kWh (1KOMMA5°, Tibber)	Higher headline price, higher total customer value; requires the hardware to be there
Flex-for-energy swap	Very low or zero commodity price in exchange for asset control (Dynamic Pulse)	Best retention weapon; only works where flex value > energy cost

Risks that sit behind the price — know these names

- **Volume risk** — the customer consumes more/less than forecast.
- **Profile / shape risk** — *when* they consume differs from the hedged shape. A dynamic tariff transfers most of this to the customer, which is precisely why it can be cheaper.
- **Imbalance risk (Ausgleichsenergie)** — deviation between the schedule and reality, settled at the imbalance price. Owned by the BKV.
- **Basis risk** — the hedge and the actual delivery point/period do not match (very relevant after the 15-min switch).

- **Credit / bad debt risk** — rises when prices spike.
- **Regulatory risk** — a levy or ruling changes the cost stack mid-contract.

TR: §41a EnWG uyarınca 2025'ten beri Almanya'da her tedarikçi dinamik tarif sunmak **zorunda** — yani ürünün varlığı artık fark değil, kalitesi fark. Fiyat kurgusu: spot fiyat + tedarikçi marjı + şebeke + vergi + KDV, artı aylık sabit ücret. Arkasındaki riskler: hacim, profil/şekil, dengeleme (Ausgleichsenergie), baz riski, tahsilat, mevzuat. Dinamik tarifi ucuz olmasının sebebi, profil riskinin müşteriye devredilmesi.

The KPI set you would actually be monitoring

KPI	Definition / why
Attach rate	% of the installed hardware base that also holds the tariff. The single most important number for a bundled model.
Time-to-supply	Days from contract signature to first supplied kWh. Driven almost entirely by Marktkommunikation quality.
Switch success rate	% of supplier switches completed without manual intervention or rejection.
Gross margin per customer per year	(energy margin + software fee + flex revenue) – (imbalance cost + service cost).
Realised vs. average price ratio	Customer's volume-weighted price ÷ simple average price. <1 means optimisation is genuinely working. This is the honest measure of HEMS value.
Flex capture	€ per kW of controllable capacity per year, split by revenue stream.
Forecast error (MAPE)	Portfolio forecast accuracy — drives imbalance cost directly.
Churn / retention	Especially churn in the first 90 days, which usually traces back to a billing or expectation failure.
Savings-promise accuracy	Promised vs. actual savings. Underrated, and a regulatory exposure.

TR: Bu tablodan en az üçünü konuşabilir olmalısın: **attach rate** (donanım tabanının kaç tane tarifi de alıyor), **time-to-supply** (sözleşmeden ilk kWh'ye kaç gün — tamamen MaKo kalitesine bağlı) ve **realised/average price ratio** (optimizasyon gerçekten işe yarıyor mu). Üçüncüsü özellikle iyi bir cevap, çünkü HEMS'in değerini dürüst ölçen metrik.

2.5 Metering — the physical precondition for everything

- **MsbG** (Messstellenbetriebsgesetz) governs metering. **iMSys** = intelligentes Messsystem: a digital meter plus a **Smart Meter Gateway (SMGW)** with certified communication. A **moderne Messeinrichtung (mME)** is digital but *not* communicating — it cannot support a dynamic tariff on its own.
- **15-minute interval values from the iMSys are what makes dynamic billing, energy sharing and §14a Module 3 possible.** No iMSys, no real dynamic product.
- The **CLS channel** in the SMGW is the standardised path for controlling devices behind the meter — the regulated alternative to a vendor cloud.
- **Since the Solarspitzengesetz (in force 25 February 2025):** new PV systems from **7 kW** must be equipped with an iMSys plus a control box by the metering point operator. Without an intelligent metering system, feed-in is capped at **60% of nominal capacity** (§9(2) EEG 2023).
- Workaround used by the market: optical meter readers (e.g. Tibber Pulse-type devices) read a modern meter locally and stream consumption to the app. Good enough for transparency and optimisation; **not** a substitute for regulated billing data.

TR: iMSys = akıllı sayaç + iletişim kutusu (SMGW). **mME** = dijital ama iletişimsiz sayaç — dinamik tarif için yetmez. 15 dakikalık veri olmadan dinamik faturalama, energy sharing ve §14a Modül 3 mümkün değil. Solarspitzenengesetz'ten (25 Şubat 2025) beri 7 kW üstü yeni PV'lerde iMSys + kontrol kutusu zorunlu; yoksa besleme %60 ile sınırlanıyor. Optik okuyucular (Tibber Pulse tipi) şeffaflık sağlar ama resmi faturalama verisi yerine geçmez.

2.6 Flexibility and the virtual power plant

A **VPP (virtuelles Kraftwerk)** pools many small assets so they behave, commercially and technically, like one dispatchable plant. Heartbeat is 1KOMMA5°'s VPP: PV, home batteries, heat pumps, EV chargers and wallboxes across >100,000 systems, now ~1 GW of shiftable load.

Revenue stacking — how the same battery earns money several ways

Stream	Mechanism	Notes
Spot arbitrage	Charge in cheap MTUs, discharge in expensive ones	Value = daily spread × usable kWh × round-trip efficiency, minus degradation
Intraday trading	Trade the pool's position after day-ahead	Also reduces the portfolio's own imbalance
Balancing markets	FCR / aFRR capacity + energy payments	Requires TSO prequalification and pooling compliance; strict availability obligations
Self-consumption	Avoid buying at full retail price (~37 ct) instead of exporting at a low feed-in rate	Usually the largest single value for a household — the "boring" one that dominates
Avoided / reduced grid fees	§14a modules; time-variable grid fee windows	Regulated, predictable, currently under-exploited
Negative-price capture	Get paid to consume when prices are below zero	Growing rapidly with solar build-out

Physical numbers to hold in your head (defensible ranges, not exact claims)

- Residential PV system: **8-12 kWp**; German specific yield ≈ **950-1,100 kWh per kWp per year**.
- Home battery: **5-15 kWh** usable; C-rate ~0.5-1; round-trip efficiency **85-92%**; ~6,000-10,000 cycles for LFP; usable depth of discharge 90-100% for LFP.
- Heat pump: **COP 3-4.5** at design point; annual **SCOP / JAZ 3.0-4.2** in German retrofits — the seasonal figure is the honest one.
- EV: **40-80 kWh** battery, home AC charging **11 kW** (3-phase); a single EV roughly doubles household consumption.
- Average German household: ~**2,500-4,000 kWh/year**; with heat pump + EV, **8,000-15,000 kWh/year**.
- §14a curtailment floor: a controllable device may be dimmed but at least **4.2 kW** must remain available.

Always present these as *ranges with the assumption stated*. In energy, a confident single number that is wrong destroys credibility faster than an honest range.

TR: VPP = binlerce küçük varlığı tek bir santral gibi yönetmek. Aynı batarya birkaç yoldan para kazanır: spot arbitraj, intraday, dengeleme piyasası, öz-tüketim, şebeke bedeli optimizasyonu, negatif fiyat yakalama. Ev için en büyük kalem genelde sıkıcı olanı: öz-tüketim. Fiziksel sayıları **aralık** olarak ve varsayımını söyleyerek ver — enerjide yanlış tek bir kesin sayı, dürüst bir aralıktan çok daha fazla itibar kaybettirir.

2.7 Direktvermarktung — the regulatory launch named in the job description

Three ways a PV system's output can be monetised in Germany:

Model	How it works	Who it suits
Einspeisevergütung (fixed feed-in tariff)	DSO pays a fixed statutory rate per kWh exported. Simple, no market exposure.	Small rooftops that want zero complexity
Marktprämienmodell (direct marketing with market premium)	A Direktvermarkter sells the output on the exchange. The grid operator pays a monthly Marktprämie = <i>anzulegender Wert</i> – <i>monthly market value</i> for that technology (floored at zero), plus a management premium. Mandatory above 100 kW.	Larger systems; the standard commercial route
Sonstige Direktvermarktung	Sell freely on the market or via PPA with no EEG premium at all. Full price exposure, full upside.	Systems that can beat the subsidy, or that want independence

Why this is suddenly a product opportunity for small rooftops

- The **Solarspitzenengesetz** substantially **simplified direct marketing below 100 kWp** — the administrative barrier that used to make it uneconomic for a 10 kWp roof has come down.
- At the same time, **§51 EEG** logic was extended: once an iMSys is installed, **no feed-in remuneration is paid for hours when the exchange price is negative**. Fixed feed-in is therefore getting structurally worse.
- Result: for a household with iMSys + battery + HEMS, "sell my surplus intelligently" starts to beat "collect a shrinking fixed rate". That is a new product line, and it is what the JD means by "regulatory product launches (e.g. Direktvermarktung)".

What the Direktvermarkter actually has to do — the operational reality

1. **Forecast** generation per 15-minute interval (weather models + system-specific model).
2. **Bid** into day-ahead by 12:00 CET; correct on intraday as forecasts update.
3. **Register** the asset with the grid operator and assign it to a **Bilanzkreis**.
4. **Submit schedules (Fahrpläne)** to the TSO — reported in defined increments.
5. **Carry the imbalance risk**: every forecast error is settled as **Ausgleichsenergie** at the imbalance price, which can be brutal in tight hours.
6. **Settle and invoice** the plant owner, and claim the market premium.

The product question you should be able to frame — and you would be unusually strong if you framed it unprompted:

"For a 10 kWp rooftop with a battery, at what monthly average market value does Direktvermarktung beat the fixed feed-in tariff, once you subtract the direct marketer's fee, the imbalance cost from forecast error, and the hours where negative prices zero out the alternative anyway? And do we pass the imbalance risk to the customer or absorb it in the portfolio?"

That single question demonstrates you understand the economics, the operations *and* the customer-experience trade-off at once. Aggregating thousands of small rooftops is exactly how a company like 1KOMMA5° makes that maths work: portfolio effects cancel individual forecast errors.

TR: Üç model: sabit besleme tarifiesi (Einspeisevergütung), piyasa primi modeli (Marktprämie = referans değer – aylık piyasa değeri), ve primsiz serbest satış. Solarspitzenengesetz 100 kWp altı için direkt satışı **basitleştirdi**; aynı zamanda iMSys varsa negatif fiyat saatlerinde besleme ödemesi **kesiliyor**. Yani sabit tarife kötüleşiyor, direkt satış cazipleşiyor — ilanda geçen "Direktvermarktung" ürünü tam olarak bu.

Operasyonel gerçek: 15 dakikalık üretim tahmini, gün-öncesi teklif, denge grubu, TSO'ya program bildirimi ve tahmin hatasının dengeleme maliyeti. Binlerce küçük çatıyı havuzlamak bu matematiği çalıştırıyor, çünkü portföyde hatalar birbirini götürüyor.

2.8 §14a EnWG — grid-side control and time-variable grid fees

- **Since 1 January 2024**, DSOs **must** connect new controllable consumption devices — heat pumps, wallboxes, home batteries — and in exchange may curtail them (down to a 4.2 kW floor) when the local grid is stressed. Connection can no longer be refused for grid reasons.
- In return, the customer chooses one of three modules:
 - **Module 1** — flat annual grid-fee discount, roughly **€110-190/year**. No second meter, simplest billing.
 - **Module 2** — percentage reduction on the grid-fee energy component; requires a **separate meter**; worthwhile for heavy users.
 - **Module 3** — **time-variable grid fees** (available since April 2025): three price windows per day; requires a **Smart Meter Gateway**. Combinable with Module 1.
- **From mid-2026** a revised BNetzA determination applies with stricter requirements, and grid-oriented control shifts from a voluntary option to a precondition for the DSO to connect controllable loads promptly.

Why the tariff team cares: Module 3 + a dynamic tariff creates a **two-signal optimisation problem** — the HEMS must minimise (spot price + time-variable grid fee) jointly, and those two signals do not always point the same way. Cheap spot hours can coincide with expensive grid windows. Solving that well is a genuine, currently unsolved product differentiator, and it is a first-class thing to raise in an interview.

TR: §14a: 2024'ten beri şebeke işletmecisi ısı pompası, wallbox ve bataryayı bağlamak **zorunda**; karşılığında sıkışma anında kısılabiliyor (en az 4,2 kW kalmak şartıyla). Müşteri üç modülden birini seçiyor: sabit indirim (~110-190 €/yıl), yüzdesel indirim (ayrı sayaç şart), veya **zaman değişkenli şebeke bedeli** (SMGW şart). 2026 ortasından itibaren BNetzA'nın yeni kararıyla kurallar sıkılaşıyor. Ürün açısından kritik nokta: Modül 3 + dinamik tarif = **iki sinyalli optimizasyon**; ucuz borsa saati pahalı şebeke penceresine denk gelebilir. Bunu iyi çözmek gerçek bir farklılaşma.

2.9 Energy sharing — a brand-new product surface (live 1 June 2026)

- **§42c EnWG** made energy sharing legally possible in Germany **from 1 June 2026**: sharing self-generated renewable electricity with other people or companies **through the public distribution grid**.
- Difference from existing models: **Mieterstrom** and **gemeinschaftliche Gebäudeversorgung (GGV)** stay inside one building; energy sharing crosses buildings — streets, blocks, neighbourhoods.
- **Scope:** initially within the same DSO settlement area; **from June 2028** extendable to neighbouring grid areas within the same control zone.
- **Requirements: iMSys on both sides**, a joint-use agreement, and each participant still needs a separate supply contract for residual power.
- **Reality check:** all DSOs are legally obliged to enable it from June 2026, but several have already signalled that technical implementation will come later. Expect a messy rollout — which for a well-organised supplier is an advantage, not a problem.

This is only weeks old as a live regime. If you can describe it accurately, you are current with the market — and "what should our energy-sharing product look like, and which DSO areas are actually ready?" is a perfect intern initiative.

TR: 1 Haziran 2026'da yürürlüğe girdi (§42c EnWG): kendi ürettiğin yenilenebilir elektriği **kamu şebekesi üzerinden** başkalarıyla paylaşabiliyorsun. Mieterstrom/GGV tek binayla sınırlıydı; bu ise mahalle ölçeğinde. Şart: iki tarafta da akıllı sayaç + ortak kullanım sözleşmesi + artık tüketim için ayrı tedarik sözleşmesi. Başlangıçta aynı şebeke bölgesi içinde, 2028'den itibaren komşu bölgelere. Birçok şebeke işletmecisi teknik olarak hazır değil — yani düzenli çalışan bir tedarikçi için fırsat. Bu konuyu doğru anlatmak "piyasayı güncel takip ediyor" demenin en net yolu.

2.10 Marktkommunikation — the section that will set you apart

The JD says: "Dive into our market communication workflows, identify inefficiencies, and drive initiatives that make our energy retail engine faster and more reliable." Here is what that sentence actually means.

What Marktkommunikation (MaKo) is

Standardised, machine-to-machine business-document exchange between all market partners in the German energy sector — supplier, DSO, metering operator, balance responsible party. It uses **EDIFACT** (the UN standard, in a BDEW-defined energy subset), exchanged over defined secure transport. Every customer relationship in German energy retail is created, changed and ended by these messages. When they fail, the customer is not supplied, not billed, or billed wrongly.

The rulebooks

Rulebook	Full name	Covers
GPKE	Geschäftsprozesse zur Kundenbelieferung mit Elektrizität	Supplier switching, move-in / move-out, master-data maintenance — the core retail processes
WiM	Wechselprozesse im Messwesen	Changing the metering point operator, meter operations
MaBiS	Marktregeln für die Bilanzkreisabrechnung Strom	Balance-group settlement: aggregating volumes and settling them between market participants
GeLi Gas	Geschäftsprozesse Lieferantenwechsel Gas	The gas equivalent of GPKE

The identifiers — get these right or nothing works

- **MaLo-ID** (Marktklokations-ID) — an **11-digit** number identifying a *market location*: a point where energy is consumed or fed in. This is the central identifier of the German electricity and gas market; supplier switches, balance-group assignment, meter-value transmission and billing all key off it.
- **MeLo-ID** (Messlokations-ID) — identifies the *metering location* (the physical meter). One market location can relate to several metering locations, and vice versa. Confusing the two is a classic, expensive error.
- **Bilanzkreis + EIC codes** — the balancing group the location is assigned to, and the standard market-participant codes.

The processes and the messages

Process	Typical message types
Lieferantenwechsel (supplier switch)	UTILMD (master data / registration), acknowledged by CONTRL / APERAK
Einzug / Auszug (move in / move out)	UTILMD
Stammdatenänderung (master-data change)	UTILMD
Meter readings / interval values	MSCONS
Invoicing between market partners	INVOIC , remittance advice REMADV
Balance-group settlement	MaBiS aggregation messages
Errors and rejections	APERAK (application error), CONTRL (syntax acknowledgement)

Why it breaks — and therefore what "improving MaKo" means

Failure modes, in rough order of frequency:

- Wrong or missing **MaLo-ID**, or a MaLo/MeLo mix-up → registration rejected.
- Mismatched master data (name, address, meter number) between supplier and DSO → rejection.
- **Deadlines missed**: GPKE processes have statutory windows; a late message restarts the process and pushes the supply start date.
- Format/version drift after a BNetzA rulebook update (these change on fixed dates and everyone has to migrate).
- Silent failures — a rejection nobody picked up, so the customer sits in limbo for weeks.
- Each of those lands as a manual clearing case, a support ticket, a delayed first invoice, and eventually churn.

How to turn this into a credible intern initiative — this is your best single answer to "what would you work on?":

1. Instrument the funnel: contract signed → UTILMD sent → acknowledged → supply start → first invoice. Measure conversion and elapsed time at each step.
2. Classify every rejection by message type × error code × DSO. Almost always a small number of causes produce most of the volume.
3. Quantify the cost: manual minutes per case × case volume, plus delayed-revenue days, plus first-90-day churn attributable to a failed start.
4. Fix the top causes at the source — validation at the point of data capture beats clearing downstream. Propose which checks move into the signup flow.
5. Report one number to the business: **switch success rate** and **median time-to-supply**, by market and by DSO.

TR: Marktkommunikation, Alman enerji piyasasında tedarikçi-şebeke-sayaç işletmecisi arasındaki standart makine-makine mesajlaşması (EDIFACT). Kural kitapları: **GPKE** (tedarikçi değişimi, taşınma, ana veri), **WiM** (sayaç), **MaBiS** (denge grubu mutabakatı). Kimlikler: **MaLo-ID** (11 haneli, tüketim/besleme noktası) ve **MeLo-ID** (fiziksel sayaç) — ikisini karıştırmak klasik ve pahalı hata. Mesajlar: **UTILMD** (kayıt/ana veri), **MSCONS** (sayaç değerleri), **APERAK/CONTRL** (hata/onay), **INVOIC**. Bozulma sebepleri: yanlış MaLo, uyuşmayan ana veri, kaçırılan yasal süreler, format sürüm değişimi, fark edilmemiş red mesajları. Sonuç: elle düzeltme, gecikmiş ilk fatura, müşteri kaybı. "MaKo'yu iyileştirmek" = bu hunideki her adımı ölçmek, red sebeplerini sınıflamak, maliyetini paraya çevirmek ve doğrulamayı kayıt anına taşımak. Görüşmede "ne üzerinde çalışırdın?" sorusuna en güçlü cevap bu.

3. Product management craft in a regulated energy business — 25%

3.1 What "own a well-scoped initiative" means in practice

Six steps. Interns who skip steps 1 and 6 are the ones who do not get the full-time offer.

1. **Frame the problem as a number.** Not "our switching is slow" but "median time-to-supply is 34 days versus a 21-day target; each extra day costs X in delayed revenue and Y in churn risk."
2. **Get the data yourself.** Do not wait for a dashboard. Pull it, clean it, sanity-check the total against a known figure.
3. **Produce 2-3 real options** with cost, effort, risk and a recommendation. One option must always be "do nothing" with its cost stated.
4. **Pressure-test with the people who own the consequences** — trading for price risk, legal for compliance, ops for workload, engineering for feasibility.
5. **Implement in the smallest shippable slice.** One market, one segment, one DSO.
6. **Measure and publish the result**, including when it did not work. Reporting a failed experiment honestly buys more trust than a vague success.

TR: "Kendi inisiyatifiğini yürütmek" şu demek: problemi **sayıya çevir** → veriyi kendin çek → maliyet/efor/risk ile 2-3 seçenek sun (biri her zaman "hiçbir şey yapmama" ve onun maliyeti) → sonucu taşıyacak ekiplerle test et → en küçük dilimle uygula → ölçüp yayınla, başarısız olduysa da. 1. ve 6. adımı atlayan stajyerler tam zamanlı teklif almıyor.

3.2 The skill that differentiates this role: regulation → product translation

Use this five-column pattern every time a law or ruling lands. It works for §14a, §42c, Solarspitzengesetz, anything:

1. Legal text	2. Obligation	3. Who is affected	4. System change	5. Customer-facing change
§42c EnWG, in force 1 Jun 2026	DSOs must enable energy sharing; participants need iMSys + joint-use agreement	Customers with PV + iMSys in a DSO area that is actually ready	New contract type, allocation logic per 15- min MTU, billing split, MaKo registration path	New product tier, eligibility checker, transparent allocation statement in the app

Then add the two columns that make it a decision rather than an analysis: **deadline** and **owner**.

TR: Bu rolün ayırt edici becerisi: mevzuatı ürüne çevirmek. Beş kolon: yasa metni → doğan yükümlülük → kimi etkiliyor → hangi sistem değişikliği → müşteriye ne olarak görünüyor. Sonuna **tarih** ve **sahip** kolonlarını ekle; analiz böylece karara dönüşür. Bu şablonu görüşmede kullanmak çok iyi durur.

3.3 Prioritisation in a regulated product

- **Compliance deadlines come first** and are non-negotiable. In energy, "we chose not to ship it" is not available if the law says June.
- For everything else, a simple **RICE** or impact/effort view is enough. Do not over-engineer frameworks in an interview.
- State the **opportunity cost** explicitly: what are we not doing because we are doing this?
- Watch for the trap: regulatory work consumes all capacity and no discretionary improvement ever ships. A good PM protects a slice for customer-experience work.

3.4 The unit-economics model you should be able to build in Google Sheets in 30 minutes

Sheets fluency is a hard requirement in the JD ("Google Sheets a must"). Build this once, from scratch, before the interview — the structure matters more than the numbers:

Block	Rows
Inputs (one clearly marked block)	Annual consumption kWh · PV kWp · specific yield · battery kWh · round-trip efficiency · spot average ct/kWh · supplier markup · base fee €/month · grid fee ct/kWh · levies+tax ct/kWh · VAT · CAC · churn %
Volume logic	PV generation · self-consumption share · grid import · export volume · shifted volume
Revenue	Energy margin (markup × import) · base fee × 12 · software fee · flex revenue €/kW/yr × controllable kW
Cost	Imbalance cost · service & support cost per customer · payment/bad debt · metering cost
Outputs	Gross margin per customer per year · payback on CAC (months) · LTV at given churn · LTV/CAC
Sensitivity	Two-way data table: spot average × churn → LTV/CAC. State which input the answer is most sensitive to.

Discipline rules: no hardcoded numbers inside formulas, every assumption on the input block with a source note, units in every label, and one sentence at the top saying what the model is for.

TR: İlan "Google Sheets şart" diyor. Görüşmeden önce bu modeli bir kez sıfırdan kur: girdiler tek blokta, hacim mantığı, gelir, maliyet, çıktı (müşteri başı yıllık brüt marj, CAC geri dönüşü, LTV/CAC) ve iki yönlü duyarlılık tablosu. Kural: formül içine sabit sayı gömme, her varsayımın kaynağını yaz, her etikette birim olsun. Sayılardan çok **yapı** değerlendirilir.

3.5 Communication habits that get an intern taken seriously

- **Answer first, then explain.** Recommendation in sentence one, reasoning after.
- **Quantify or flag.** Every claim is either a number with a source or explicitly labelled an assumption.
- **Say "I don't know" cleanly,** then say how you would find out and by when. In an energy interview this is a strength, because the domain is genuinely large and pretending is transparent.
- **Disagree with the mechanism, not the person:** "I'd expect that to break at step X because Y — have we tested that?"
- **Never present a number you cannot reproduce.** You will be asked where it came from.

TR: Önce cevap, sonra gerekçe. Her iddia ya kaynaklı bir sayı ya da açıkça "varsayım" etiketli. **"Bilmiyorum"** demeyi öğren — nasıl öğreneceğini ve ne zaman söyleyerek. Enerjide bu güç göstergesi, çünkü alan çok geniş ve bilmiş görünmek anında anlaşılıyor. Üretmediğin bir sayıyı asla sunma; kaynağı sorulur.

4. Data & tooling — 15%, but where you already have an edge

The JD asks for "analytical and structured thinker, comfortable with data and spreadsheets (Google Sheets a must, SQL or Python a plus)". So data is a supporting skill here — but it is *your* supporting skill, and it is how you will produce work faster than the other interns.

4.1 Google Sheets, to a professional standard

- `QUERY()` — SQL-like filtering and aggregation inside Sheets. The single highest-leverage function; learn it properly.
- `SUMIFS` / `COUNTIFS` / `AVERAGEIFS`, `XLOOKUP` (or `INDEX/MATCH`), `ARRAYFORMULA`, `UNIQUE`, `FILTER`, `SPLIT`.
- Pivot tables with calculated fields; two-way **data tables** for sensitivity; **named ranges**; data validation; conditional formatting for threshold flags.
- `IMPORTDATA` / `IMPORTJSON` -style pulls for price feeds if a live sheet is needed.
- Model hygiene: inputs / calculations / outputs on separate blocks or tabs; units in labels; a "read me" cell; no magic numbers.

4.2 SQL you will actually use

- `SELECT` / `WHERE` / `GROUP BY` / `HAVING` / `ORDER BY`; all four join types and when an outer join hides a data-quality bug.
- **Window functions** — the ones that matter for interval energy data: `LAG()` / `LEAD()` for deltas between consecutive intervals, `SUM()` `OVER (PARTITION BY ... ORDER BY ...)` for cumulative volume, `ROW_NUMBER()` for deduplication, `AVG()` `OVER` for rolling baselines.
- `DATE_TRUNC` / bucketing to 15-minute, hourly, daily granularity; calendar joins so missing intervals show as gaps instead of vanishing.
- **CTEs** (`WITH`) to keep multi-step logic readable — this is a communication skill as much as a technical one.
- A cohort/retention query: customers grouped by signup month, activity by month since signup.

4.3 Time-series realities that cause real bugs

- **96 intervals per day, 35,040 per year** at 15-minute resolution. Per meter. Volume grows fast.
- **Daylight saving time**: in the CET/CEST zone one day per year has **23 hours** and one has **25**. Every naive $\times 96$ assumption breaks twice a year. Store and compute in **UTC**, display in local time.
- **Missing and substitute values** (*Ersatzwerte*): meter data arrives late, incomplete, or estimated. Never silently treat missing as zero — a missing interval and a zero-consumption interval mean completely different things commercially.
- **kW vs kWh**: power vs energy. At 15-minute resolution, energy = average power $\times$ 0.25 h. Mixing these up is the most common analytical error in the industry.
- **Sign conventions**: import positive / export negative, or two separate registers? Get this defined before you compute anything.
- **Restatements**: meter values get corrected after the fact, so yesterday's report legitimately changes. Version your numbers.

TR: Enerji verisinde gerçek hatalar buradan çıkıyor: 15 dakikalık çözünürlükte günde 96, yılda 35.040 aralık; yaz saati yüzünden yılda bir gün 23, bir gün 25 saat (UTC'de sakla, yerel saatte göster); eksik veri sıfır demek **değil** (Ersatzwerte); kW ile kWh'yi karıştırma (15 dk'da enerji = ortalama güç × 0,25); işaret konvansiyonunu baştan tanımla; sayaç değerleri sonradan düzeltilir, yani dünkü rapor meşru şekilde değişir.

4.4 Python — and the one analysis worth actually doing before the interview

pandas essentials: `read_csv`, `to_datetime`, `set_index`, `resample('15min')`, `groupby().agg()`, `merge`, `pivot_table`, `rolling()`, `tz_localize` / `tz_convert`.

The single most impressive thing you could bring: a small **tariff backtest**. Take a year of German day-ahead prices and one load profile. Compute: (a) the bill under a fixed tariff, (b) the bill under a dynamic tariff with no behaviour change, (c) the bill with a battery shifting N kWh/day into the cheapest quarter-hours. Report the **realised/average price ratio** and the euro delta — and state your assumptions and their sensitivity.

This is one afternoon of work. It demonstrates energy understanding, data skill, product judgement and honesty about assumptions simultaneously. Almost no internship candidate does it.

4.5 Data sources and APIs to know by name

Source	What you get
ENTSO-E Transparency Platform	Pan-European day-ahead prices, load, generation by type, cross-border flows. The canonical free source; free API key.
EPEX SPOT	The exchange itself — day-ahead and intraday products, market rules, the 15-minute MTU documentation.
SMARD (BNetzA)	Official German market data portal — generation, consumption, prices, balancing. Free downloads, good for sanity checks.
Energy-Charts (Fraunhofer ISE)	Best free visualisations of German generation and prices; excellent for building intuition quickly.
Agora Energiewende / Agorameter	Policy-grade analysis and data on the German transition.
aWATTar / Tibber public price APIs	Retail-facing hourly/quarter-hourly price endpoints — how competitors expose prices to customers and devices.
Marktstammdatenregister (MaStR)	Official register of every generation asset in Germany. Useful for market sizing.
BNetzA / BDEW publications	Rulebook changes, determinations, price statistics. The source of your deadlines.

4.6 How the data stack in this kind of company usually looks

Device telemetry → event streaming (Kafka / MQTT ingestion) → cloud warehouse (Snowflake / BigQuery / Databricks) → transformation layer (dbt) → BI (Looker / Metabase / Power BI) plus reverse-ETL into operational tools. Market data and MaKo messages land as separate pipelines and get joined on **MaLo-ID**.

Your Fabric experience maps one-to-one — say it in their vocabulary: bronze/silver/gold = raw / staging / marts; Lakehouse Delta tables = warehouse tables; notebooks = transformation jobs; DAX measures = the semantic/metrics layer; EventStream + KQL = streaming ingestion + hot

path. You are not learning a new concept, you are learning a different vendor's word for it. Say exactly that — it signals transferability instead of tool loyalty.

TR: Fabric bilgin birebir karşılık buluyor: bronze/silver/gold = raw/staging/marts, Lakehouse = veri ambarı, notebook = dönüşüm işi, DAX ölçüleri = metrik katmanı, EventStream+KQL = akış yutma. Görüşmede tam olarak bunu söyle: "kavramlar aynı, sadece satıcının kelimeleri farklı." Bu, araç bağımlılığı değil transfer edilebilirlik sinyali verir.

5. Your own bridge — how EnergyLens maps to this job

You have an unusual asset for an internship candidate: you have already built the analytical half of this exact problem domain. Map it explicitly rather than describing it generally.

What the JD asks for	What you have already built
Work on dynamic electricity tariff products	A daily pipeline ingesting day-ahead prices (EPEX-based) with country/hour granularity, fallback logic and CO ₂ intensity, feeding downstream ROI calculations
Understand flexibility and asset economics	A battery dispatch and simulation module with strategy comparison, ROI/payback and an explicit honest split of measured vs. modelled vs. prospective results
Analyse and improve process/data reliability	A five-protocol IoT ingestion layer (BACnet/IP, Modbus TCP, MQTT, REST, plus edge agent) normalising heterogeneous device data to standard units and a common schema
Monitor KPIs and present findings	A full KPI/metrics layer with 200+ measures over a medallion architecture, plus a 10-page decision-support report and a live web application
Regulatory product launches	Compliance modules built against real regimes: EU Battery Regulation 2023/1542, ESRS-E1-aligned reporting, German CO ₂ KostAufG, EPBD/MEPS logic
Ownership with minimal handholding	You scoped, architected, built, deployed and documented the whole platform solo, and made the strategic call to build production-grade rather than MVP

Say the limitation before they find it. EnergyLens runs on synthetic and estimated data; there is no completed customer pilot yet. State that plainly and early. Two reasons: it is true, and in an energy interview an unprompted honest caveat is read as domain maturity. The version that damages you is being caught claiming validated savings you cannot evidence.

Frame it as: *"I built the full analytics stack end to end on synthetic and public data. What I have not yet done is validate the savings model against a real metered building — which is exactly the part I would learn fastest inside a company with 100,000 connected systems."*

TR: Bu tablo senin en güçlü kozun: ilanın her satırına karşılık gelen somut bir şey inşa etmişsin. Ama **sınırı sen söyle:** EnergyLens sentetik/tahmini veriyle çalışıyor, tamamlanmış bir pilot yok. Bunu erken ve net söylemek enerji sektöründe olgunluk sinyali; yakalanmak ise itibar kaybı. Kalıp: "tüm analitik yığının uçtan uca kurdum, henüz yapmadığım şey modeli gerçek sayaç verisiyle doğrulamak — ve bu tam olarak 100.000 bağlı sistemi olan bir şirkette en hızlı öğrenebileceğim şey."

6. Study plan — what to do with the time you have

6.1 Before the call (highest return per hour, in this order)

#	Task	Why it is first
1	Scan the work permit, Zusatzblatt and residence permit; write down your permitted weekly hours	They asked for it in the invitation. Being unprepared here is the only avoidable disqualifier.
2	Learn Section 7 glossary cold — German terms with their English meaning	You will hear Netzentgelt, Bilanzkreis, Direktvermarktung, Marktkommunikation, Lieferantenwechsel in the call. Recognition is 80% of the confidence.
3	Be able to recite the retail price stack (2.3) and the "only ~40% is dynamic" conclusion	It is the fastest proof that you think in numbers, not slogans.
4	Prepare the Marktkommunikation initiative from 2.10 as your "what would you work on" answer	Directly from the JD; almost no candidate can do it.
5	Prepare a 90-second EnergyLens story with the honest caveat built in	Your differentiator, delivered with credibility intact.
6	Prepare your German answer: current level, evidence, and what you are doing about it	Do not let this question find you unprepared.
7	Spend 30 minutes on Energy-Charts and SMARD looking at a real week of prices	Turns abstract knowledge into visual intuition. You will speak differently afterwards.
8	Write three real questions for them (see 6.3)	Questions reveal level more than answers do.

6.2 Two-week deepening plan (also the right first-month plan on the job)

Days	Focus	Output you produce
1-2	Market roles + wholesale sequence (2.1, 2.2)	One page: who does what, drawn from memory
3-4	Retail price stack + dynamic tariff mechanics (2.3, 2.4)	Sheets model: bill under fixed vs. dynamic tariff
5-6	Metering + §14a (2.5, 2.8)	Decision table: which module suits which customer, and why
7-8	Direktvermarktung economics (2.7)	Break-even analysis: DV vs. fixed feed-in for a 10 kWp roof
9-10	Marktkommunikation processes (2.10)	Process map: contract → first invoice, with failure points marked
11-12	Flexibility & VPP revenue stacking (2.6)	Python backtest: dynamic tariff with and without battery shifting

13-14	Energy sharing + competitor teardown (2.9, 1.4)	One-pager: what our energy-sharing product should be, and which competitor is closest
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Every row produces an artefact. Reading without producing does not create confidence — producing does. Six of these seven artefacts are also directly reusable as work samples.

6.3 Questions worth asking them

- "How is ownership split between the global tariff product team and the local market teams — where does a Germany-specific regulatory change get decided?"
- "With Module 3 time-variable grid fees, are you optimising jointly against spot and grid fee yet, or is spot still the dominant signal?"
- "On Direktvermarktung for small rooftops — is the plan to absorb imbalance risk in the portfolio or pass it to the customer?"
- "What does the switching funnel look like today — where do you lose the most days between contract and first supplied kWh?"
- "Is the Swedish Dynamic Pulse logic transferable to Germany given the non-commodity share of the bill, or does it become a discount rather than a zero?"
- "What separates an intern who converts to full-time from one who does not, in this team specifically?"

Pick two or three. Each of these is only askable by someone who has understood the business, which is the point.

TR: Sıralama önemli: (1) belgeler, (2) Almanca terim sözlüğü, (3) fiyat yapısı ve "sadece %40'ı dinamik" sonucu, (4) MaKo inisiyatifi cevabı, (5) EnergyLens hikâyesi + dürüst çekince, (6) Almanca seviyesi cevabı, (7) gerçek fiyat grafiklerine bak, (8) soruların. İki haftalık planın her satırı bir **çıktı** üretiyor — okumak güven vermiyor, üretmek veriyor. Soru sormak seviyeni cevaplardan daha çok gösterir; listeden 2-3 tanesini seç.

7. Glossary — German / English / Turkish

Learn the left column by recognition. This is the highest-value page in the document for working in a German energy company.

German	English	Türkçe
Energiewende	Energy transition	Enerji dönüşümü
Stromlieferant / Versorger	Electricity supplier	Elektrik tedarikçisi
Grundversorger / Grundversorgung	Default supplier / basic supply	Son kaynak tedarikçi (pahalı varsayılan tarife)
Übertragungsnetzbetreiber (ÜNB)	Transmission system operator (TSO)	İletim şebekesi işletmecisi
Verteilnetzbetreiber (VNB)	Distribution system operator (DSO)	Dağıtım şebekesi işletmecisi
Netzentgelt / Netzgebühren	Grid fee / network charge	Şebeke kullanım bedeli
Netzanschluss	Grid connection	Şebeke bağlantısı
Bilanzkreis	Balancing group	Denge grubu
Bilanzkreisverantwortlicher (BKV)	Balance responsible party	Denge grubu sorumlusu
Ausgleichsenergie	Balancing / imbalance energy	Dengeleme enerjisi (tahmin sapmasının maliyeti)
Regelleistung / Regelenergie	Balancing reserve / control power	Yedek kapasite piyasası
Primär-/ Sekundärregelleistung	FCR / aFRR	Primer / sekonder frekans kontrolü
Minutenreserve	mFRR / tertiary reserve	Dakika yedeği
Redispatch	Redispatch	Şebeke sıkışması için üretim kaydırma
Fahrplan	Schedule / nomination	Program bildirimi
Börsenstrompreis / Spotpreis	Exchange / spot price	Borsa elektrik fiyatı
Day-Ahead-Auktion	Day-ahead auction	Gün öncesi ihale
Intraday-Handel	Intraday trading	Gün içi ticaret
Marktzeiteinheit (MTU), Viertelstunde	Market time unit, quarter-hour	Piyasa zaman birimi, çeyrek saat
Merit Order	Merit order	Maliyet sıralı devreye alma
Negative Preise	Negative prices	Negatif fiyatlar
Beschaffung	Procurement / sourcing	Tedarik (enerji alımı)
Absicherung / Hedging	Hedging	Riskten korunma
Dynamischer Stromtarif	Dynamic electricity tariff	Dinamik elektrik tarifi
Grundpreis / Arbeitspreis	Base fee / unit price	Sabit ücret / birim fiyat
Aufschlag	Markup / adder	Tedarikçi marjı

Stromsteuer	Electricity tax	Elektrik vergisi
Umlage (KWKG, §19 StromNEV, Offshore)	Levy / surcharge	Yasal harç / katkı payı
Konzessionsabgabe	Concession fee	İmtiyaz bedeli
Mehrwertsteuer (MwSt)	VAT	KDV
Zähler / Zählerstand	Meter / meter reading	Sayaç / sayaç okuması
Intelligentes Messsystem (iMSys)	Smart metering system	Akıllı sayaç sistemi
Moderne Messeinrichtung (mME)	Modern (non-communicating) meter	Dijital ama iletişimsiz sayaç
Smart-Meter-Gateway (SMGW)	Smart meter gateway	Akıllı sayaç iletişim ağ geçidi
Messstellenbetreiber (gMSB / wMSB)	Metering point operator (default / competitive)	Sayaç işletmecisi (varsayılan / rekabetçi)
Marktlotation (MaLo-ID)	Market location ID (11-digit)	Piyasa noktası kimliği
Messlotation (MeLo-ID)	Metering location ID	Ölçüm noktası kimliği
Marktkommunikation (MaKo)	Market communication	Piyasa katılımcıları arası standart mesajlaşma
Lieferantenwechsel	Supplier switch	Tedarikçi değişikliği
Einzug / Auszug	Move-in / move-out	Taşınma giriş / çıkış süreci
Stammdaten	Master data	Ana veri
GPKE / WiM / MaBiS	Retail / metering / settlement rulebooks	Süreç kural kitapları
Direktvermarktung	Direct marketing of generation	Üretimin piyasada doğrudan satışı
Direktvermarkter	Direct marketer	Direkt satış aracı
Marktprämie / Managementprämie	Market premium / management premium	Piyasa primi / yönetim primi
Anzulegender Wert	Applicable (reference) value	Referans destek değeri
Monatsmarktwert	Monthly market value	Aylık piyasa değeri
Einspeisevergütung	Feed-in tariff	Sabit besleme tarifesi
Einspeisung / Eigenverbrauch	Feed-in / self-consumption	Şebekeye verme / öz tüketim
Volleinspeisung / Überschusseinspeisung	Full feed-in / surplus feed-in	Tam besleme / fazla besleme
Einspeisebegrenzung (60%-Regel)	Feed-in cap (60% rule)	Besleme sınırlaması
Steuerbare Verbrauchseinrichtung	Controllable consumption device	Kontrol edilebilir tüketim cihazı
Netzorienteerte Steuerung	Grid-oriented control / curtailment	Şebeke odaklı kısıtlama
Zeitvariable Netzentgelte	Time-variable grid fees	Zaman değişkenli şebeke bedeli
Energy Sharing (§42c EnWG)	Energy sharing	Enerji paylaşımı (komşuyla)

Mieterstrom	Tenant electricity	Kiracı elektriği modeli
Gemeinschaftliche Gebäudeversorgung (GGV)	Collective building supply	Bina içi ortak tedarik
Virtuelles Kraftwerk	Virtual power plant (VPP)	Sanal santral
Flexibilität / Lastverschiebung	Flexibility / load shifting	Esneklik / yük kaydırma
Wärmepumpe (JAZ)	Heat pump (seasonal performance factor)	Isı pompası (yıllık verim katsayısı)
Wallbox / Ladepunkt	Wallbox / charging point	Ev şarj ünitesi
Speicher / Batteriespeicher	Storage / battery storage	Depolama / batarya
Erzeugungsanlage	Generation asset	Üretim tesisi
EnWG / EEG / MsbG	Energy Industry Act / Renewables Act / Metering Act	Enerji Piyasası / Yenilenebilir / Sayaç kanunları
Bundesnetzagentur (BNetzA)	Federal Network Agency (regulator)	Federal düzenleyici kurum
Beschluss / Festlegung	Regulatory determination / ruling	Düzenleyici karar
Solarspitzen-gesetz	"Solar peak" act	Güneş tepe üretimi yasası
Marktstammdatenregister (MaStR)	Market master data register	Resmî tesis kayıt sistemi

8. Honest risk register

Risk	Severity	How to handle it
German language	High	The JD asks for excellent German. Do not bluff. State your real level, note that you can read regulatory German (EnWG/EEG/BNetzA terminology — true after Section 7), commit to a concrete improvement path, and point out that MaKo and price analysis are largely structured-data work. Ask directly what the day-to-day language of the team is — some 1KOMMA5° global product teams operate in English.
Work permit / Zusatzblatt	High	They asked in writing. Have scans ready before the call and know your permitted hours. This is administrative, not a judgement on you — but arriving unprepared reads as carelessness.
6 months, ideally full-time	Medium	Check this against your MSc obligations and your freelance/EnergyLens commitments <i>before</i> the call, and give a clear answer. Ambiguity here worries hiring managers more than a "no".
Profile fit vs. business/econ students	Medium	You will be compared against candidates with consulting internships. Your counter is not "I also know business" — it is that you can build the analysis they can only request. Lead with the tariff backtest and the unit-economics model.
Over-claiming on EnergyLens	Medium	Synthetic data, no completed pilot. Say it first, every time.
Being seen as "the data person"	Low-Medium	This is a product role. Whenever you describe a technical capability, close the loop with the decision it enabled — otherwise you get filed as an analyst, not a future PM.

TR: Gerçek riskler sıralı: (1) **Almanca** — bluff yapma, gerçek seviyeyi + mevzuat Almancasını okuyabildiğini + somut planı söyle ve ekibin günlük dilini sor (1KOMMA5°'nin global ürün ekipleri sıklıkla İngilizce çalışıyor). (2) **Çalışma izni** — belgeleri görüşmeden önce hazırla, izinli saatini bil. (3) **6 ay tam zamanlı** — yüksek lisans ve freelance yükünü karşılamasını önceden netleştir; belirsizlik "hayır"dan daha çok endişe yaratır. (4) **Profil karşılaştırması** — danışmanlık stajı yapmış işletme öğrencileriyle yarışyorsun; kozun "ben de iş bilirim" değil, "onların sadece talep edebildiği analizi ben kurabiliyorum". (5) EnergyLens'te abartma. (6) "Veri adamı" damgası yememek için her teknik yetenek anlatımını, sağladığı **karar** ile kapat.

Closing note

Confidence in this domain does not come from knowing everything — the German energy market is genuinely too large for that, and everyone in the room knows it. It comes from three things: knowing the **structure** (who does what, in what order, under which rule), knowing your **numbers** (and their sources and their sensitivity), and being **precise about what you do not know**. You already have the hardest part, which is the ability to build. This document is the vocabulary and the map to put around it.

TR: Bu alanda güven her şeyi bilmekten gelmez — Alman enerji piyasası bunun için fazla geniş ve odadaki herkes bunu biliyor. Güven üç şeyden gelir: **yapıyı** bilmek (kim ne yapıyor, hangi sırada, hangi kurala göre), **sayılarını** bilmek (kaynağı ve duyarlılığıyla) ve **bilmediğin şeyi net söylemek**. Zor olan kısım — inşa etme becerisi — sende zaten var. Bu doküman onun etrafına giydirilecek kelime dağarcığı ve harita.

Sources used: 1KOMMA5° job posting and press releases · pv-magazine · smart-energy.com · ess-news · Sifted / Dealroom / techfundingnews (funding) · EPEX SPOT and Next Kraftwerke (15-minute day-ahead) · BNetzA-derived price composition summaries (Mainova, strom-report, EnBW) · §14a EnWG explainers (Netze BW, Vattenfall, Finanztip) · Solarspitzenengesetz explainers (metergrid, reonic, Logic Energy) · §42c energy sharing (GÖRG, DGS, metergrid) · Marktkommunikation references (Wikipedia, stromhaltig, enerchy) · dynamic tariff market comparisons 2026. Figures for 2026 are current as of August 2026 and should be re-checked before quoting externally. Physical performance figures are stated as ranges with assumptions, not guarantees.